

|                                                      |                                          |
|------------------------------------------------------|------------------------------------------|
| Program : <b>Diploma in Manufacturing Technology</b> |                                          |
| Course Code : <b>5709</b>                            | Course Title: <b>CNC Programming Lab</b> |
| Semester : <b>5</b>                                  | Credits: <b>1.5</b>                      |
| Course Category: <b>Program Elective</b>             |                                          |
| Periods per week: <b>3 (L:0, T:0, P:3)</b>           | Periods per semester: <b>45</b>          |

### Course Objectives:

- To provide the basic knowledge in part programming in CNC lathe and milling jobs
- To familiarize with basic codes in CNC programming and programming methods
- To familiarize with editing and simulation in CNC programming

### Course Prerequisites:

| Topic               | Course Code | Course Name                    | Semester |
|---------------------|-------------|--------------------------------|----------|
| Basic Science       |             | Engineering Mathematics I & II | 1, 2     |
| Basic Science       |             | Applied Physics I & II         | 1, 2     |
| Engineering Science |             | Engineering Graphics           | 1        |
| Program Core Course |             | Machine Tools                  | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| COn | Description                                  | Duration (Hours) | Cognitive Level |
|-----|----------------------------------------------|------------------|-----------------|
| CO1 | Study of CNC machines and programming codes. | 5                | Understanding   |
| CO2 | Part programming on CNC lathe                | 13               | Applying        |
| CO3 | Part programming on CNC Milling machine      | 15               | Applying        |
| CO4 | Program editing and simulation               | 9                | Applying        |
|     | Lab Exam                                     | 3                |                 |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 2   |     |     |     |     |     |     |
| CO2             | 3   | 3   |     |     |     |     |     |
| CO3             | 3   | 3   |     |     |     |     |     |
| CO4             | 3   | 3   | 3   |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline**

| Module Outcomes                                                                                                                                                                                                                             | Description                                                                  | Duration (Hours) | Cognitive Level |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                         | <b>Study of CNC machines and programming codes.</b>                          |                  |                 |
| M1.01                                                                                                                                                                                                                                       | Study of CNC Machines                                                        | 3                | Remembering     |
| M1.02                                                                                                                                                                                                                                       | Study of Programing codes                                                    | 2                | Understanding   |
| <b>Contents:</b> Introduction to Computer Numerical Control , Understand the CNC Machines and programming codes                                                                                                                             |                                                                              |                  |                 |
| CO2                                                                                                                                                                                                                                         | <b>Part programming on CNC lathe</b>                                         |                  |                 |
| M2.01                                                                                                                                                                                                                                       | Study Co ordinate System ,Position Planes , Cartesian and Polar Co ordinates | 4                | Understanding   |
| M2.02                                                                                                                                                                                                                                       | Study the various axis motion and direction, reference points                | 4                | Understanding   |
| M2.03                                                                                                                                                                                                                                       | Prepare part program for CNC lathe                                           | 5                | Applying        |
|                                                                                                                                                                                                                                             | Lab Exam – I                                                                 | 1.5              |                 |
| <b>Contents:</b> Co ordinate System, Position Planes, Cartesian and Polar Co ordinates Familiarize with the various axis motion and direction-Understand Machine Zero, Work Zero-Identify tool reference points-part programming exercises. |                                                                              |                  |                 |
| CO3                                                                                                                                                                                                                                         | <b>Part programming on CNC Milling machine</b>                               |                  |                 |
| M3.01                                                                                                                                                                                                                                       | Study Co ordinate System ,Position Planes , Cartesian and Polar Co ordinates | 5                | Understanding   |

|                                                                                                                                                                                                                                                                      |                                                                        |     |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----|---------------|
| M3.02                                                                                                                                                                                                                                                                | Study the various axis motion and direction, reference points          | 5   | Understanding |
| M3.03                                                                                                                                                                                                                                                                | Prepare part program for CNC Milling machine                           | 5   | Applying      |
| <b>Contents:</b> Co ordinate System, Position Planes, Cartesian and Polar Co ordinates<br>Familiarize with the various axis motion and direction-Understand Machine Zero, Work Zero Identify tool reference points-part programming problems on CNC milling machine. |                                                                        |     |               |
| <b>CO4</b>                                                                                                                                                                                                                                                           | <b>Program editing and simulation</b>                                  |     |               |
| M4.01                                                                                                                                                                                                                                                                | Prepare edit and simulation of Part programming on CNC lathe           | 4   | Applying      |
| M4.02                                                                                                                                                                                                                                                                | Prepare edit and simulation of Part programming on CNC Milling machine | 3   | Applying      |
| M4.03                                                                                                                                                                                                                                                                | Study to prepare Sub programmes                                        | 2   | Applying      |
|                                                                                                                                                                                                                                                                      | Lab Exam – II                                                          | 1.5 |               |
| <b>Contents:</b> Program edit-simulation-CNC lathe and milling machines-prepare sub programmes.                                                                                                                                                                      |                                                                        |     |               |

### Text / Reference

| T/R | Book Title/Author                                                                     |
|-----|---------------------------------------------------------------------------------------|
| T1  | CNC Programming by R SK Sinha – Galgotia Publication, New Delhi                       |
| R1  | Computer Numerical Control Machine – Radhakrishnan P , New Central Book Agency Agency |
| R2  | CNC Part Programming Manual KELTRAC                                                   |

### Online Resources

| Sl.No | Website Link                                                                                                                                          |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://www.cnccookbook.com/g-code-m-code-reference-list-cnc-mills/">https://www.cnccookbook.com/g-code-m-code-reference-list-cnc-mills/</a> |
| 2     | <a href="https://nptel.ac.in/content/storage2/courses/112103174/pdf/mod7.pdf">https://nptel.ac.in/content/storage2/courses/112103174/pdf/mod7.pdf</a> |